

[Lecture Note] Entering, Coding, and Analyzing Qualitative Data

GE2021 Research Technology, Spring 2025

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Learning Objectives

- 12-1 Describe how to enter, clean, and organize qualitative data.
- 12-2 Examine and code raw data.
- 12-3 Analyze qualitative data.

SHORT VIDEO INTRODUCTION

<https://www.youtube.com/watch?v=lYzhgMZii3o>

<https://www.youtube.com/watch?v=Wj45MVxHWOA>

<https://www.youtube.com/watch?v=rvMf1cbctYM>

<https://www.youtube.com/watch?v=eT-EDgwRvRU>

<https://www.youtube.com/watch?v=95hHHIW6k6Q>

Enter, Clean, and Organize Qualitative Data

- Conceptually, qualitative information is handled in the same way as quantitative information, but some adjustments must be made because the data are delivered in immediately quantifiable form.
- By its very nature, qualitative information will seldom be represented by numbers, but it can still be systematic and well organized.
- In addition to the raw information accumulated when conducting a study, it is essential to create a separate file of various records, including how many interviews, case studies, or observations were conducted, when and where the data collection took place, and other information about the participants in the study.
 - Basic information like the date, time, location, duration of the event, and participant characteristics should be logged.

Transcription

- Transcribing: the process where the recording of an interview or a focus group or any other recorded material is turned into written words.
- There are various dictation applications and software that can help with transcription process.
- A researcher needs to include everything that is heard, from grammatical errors and mispronunciations to background noise.

Memoing

- Taking notes on what you are thinking as you read and reread the transcribed interviews is called memoing, which facilitates later analysis by bringing you closer to the written material.
- The researcher relies on these memos or written impressions that were collected during the analysis to see for consistencies or repeated thoughts and ideas.
- Classification of memos is a process that happens after the memos are written in free-thinking style.
 - **Theoretical memos:** These memos relate to a theory or make connections between the narrative and theoretical concepts.
 - **Self-reflection memos:** These memos connect the narrative with past experiences or simply note how the researcher perceives the written information.
 - **Hypothetical memos:** These memos propose or predict a relationship between different parts of the narrative, though it is still too early to come to a conclusion.

Diagrams

- Diagrams are not only useful for record keeping but can also lend a systematic and organized view of the data.
- Various software programs allow researchers to create a range of diagrams, from simple to complex.

Coding Qualitative Data

- **Coding** is one way of bringing order to this written information by labelling and categorizing data.
- **A code** is a word or a few words that capture common meaning or categorization.

- The process of coding involves aggregating the text or visual data into small categories of information, seeking evidence for the code from different databases being used in a study, and then assigning a label to the code.
 - **In vivo coding:** code word per word as written in text.
 - **Preexisting codes:** codes taken from the literature.
 - **Theoretical coding:** a set of structured codes, usually from the theoretical perspective of the study, and the text is further examined with little flexibility to add other interpretations.
 - **Content analysis:** a type of qualitative analysis where frequencies of certain words used in a text are examined.
 - **Patterns:** the identification and grouping of similarities or differences of codes.
 - **Interpretation:** the process of going from simple codes to patterns and themes in the text.
 - **Themes:** broader groups into which categories of memos are gathered.

Analyzing Qualitative Data

While researchers have attempted to come up with many systematic ways to handle qualitative data, the analytical process can still be fraught with difficulty because each researcher is as unique as their study. However, qualitative analysis, when conducted correctly, should correct for those individual differences and bring out the best possible interpretation.

- **Recognizing Patterns**
 - Looking for similarities in actions, words, events, or feelings is one way of investigating qualitative work.
- **Creating Categories**
 - At the start, coding involves removing extraneous information while keeping the information of value.
 - Then, in the analysis phase, we assemble similarly coded information and organize it into larger categories so we can see how different participants approached the same topic of interest.

- **Searching for Frequency**

- Frequency plays an important role in qualitative analysis, because the repetition of a topic or concept may be meaningful.

- **Considering Diversity**

- Since diversity is part of the world we live in, our continued awareness of it will result in better studies, better investigation, and much better results and discussions.

- **Remembering the Past**

- There are many cases when past experiences, personal history, or other types of background will influence participants and impact their responses.
 - Taking into account as many elements as we can when analyzing data will result in a higher quality of data analysis and a stronger study.
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- **Thinking Theoretically**

- Considering theories while analyzing data can help researchers interpret information in a meaningful way and make necessary connections between their material and the larger theoretical context.

- **Thinking Critically**

- Critical thinking is the ability to look at something from multiple perspectives and to challenge our usual ways of thinking.

Class Activities:

- This activity will provide students with experience transcribing qualitative data. Ask each student to identify a friend or family member he or she can interview, on the phone or in-person. The purpose of the interview is to ask this person the following question, “Tell me a story about what you most proud of and why?” Ask the students to tape-record this interview using a digital tape recorder or their phone, with the permission of the participant. Be sure participants can clearly describe the purpose of this project to interview participants before conducting these interviews. Before commencing transcription, convene students in class to have a discussion about the structure for the transcripts. More specifically, discuss what type of information you will include in the transcript (e.g., date or location) and how the data will be recorded. For example, will the student transcribe verbatim (including non-verbal exchanges)? Encourage the students to download a free transcription software such as Express Scribe to facilitate the transcription process (<https://www.nch.com.au/scribe/index.html>). After each student transcribes their data have a discussion about this process and what facilitated and challenged this process.

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- This activity will allow students to practice the creation of a codebook for qualitative research. Create a document on a shared drive (e.g., Google drive) with the following question “What does the word community mean to you?” Before the classroom session, ask each student to respond to this question in one paragraph in this document on the shared drive. Once you have gathered all the responses from all the students in one document, bring the class together as a group to do the following:
 - (a) Develop a set of preexisting codes or things you hope to “hear” in the text.
 - (b) Then divide the class into groups of three to five students and have them review the document with student responses and add codes to the codebook using an in vivo coding strategy.
 - (c) Discuss what you learned from this codebook development process including what was challenging about this experience for students.

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- This activity will enable students to gain experience in coding and think about inter-rater reliability in coding. This activity builds on the activity from Question 2. Divide students into groups of two and provide each student with a copy of the Google drive with student responses to the Question “What does community mean to you?” and the final codebook. Ask each student to separately apply the codebook to the data. Students can apply codes by handwriting them in the margins of the document or use the insert comment feature in Word. After each student in a pair has independently coded the Google document, ask the students to discuss and reconcile their coding with their partner. After this process is complete, discuss challenges related to doing this work with the class as a whole.

House Keeping

Announcement

Assignment

Assignment1: Assignment Title

- **What to do**
 - A
 - B
- **Format and Requirement**
 - PDF format, < ?? pages
 - file name should be include your student id and name
 - * stuID_name_title.pdf (e.g. 1111111_ChungilChae_SelfIntroduction.pdf)
 - APA, Double-Space, No title page, 12 points, Reference section if necessary

- 152 – Chinese name in English, English name, Student ID, Class name and section (e.g. GE2021,
153 W09; MGS3001, W01)
- 154 • due date
- 155 – by Date(DAY) 11:59PM
- 156 – NO LATE SUBMISSION ALLOWED!!!!

157 **Reference**